

One Line, Four Ways to Show It

Answer Key

Grade 8 / Pre-Algebra

Quick Answers

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|---|--|
| 1. (a) y -intercept = 6, (b) slope = 4 | 6 = 19, — |
| 2. cost of a 12-mile ride = 27 | 9. (a) total after 7 months = 176, (b) months to reach 284 dollars = 13 |
| 3. (a) slope = 3, (b) y when x is 7 = 23 | 10. (a) change in y = 18, (b) slope = $\frac{3}{2}$ |
| 4. (a) table value at 4 = 24, (b) equation value at 4 = 24, — | 11. (a) litres from the table = 78, (b) litres from the equation = 78, — |
| 5. charge after 9 hours = 28 | 12. (a) table value at 4 = 14, (b) value at 20 = 62, — |
| 6. (a) y -intercept = 40, (b) slope = -3 | |
| 7. (a) slope = -2 , (b) y when x is 3 = 4 | |
| 8. (a) slope of line R = 2, (b) equation value at | |
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What is verified

8 of 12 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 4, 8, 11, 12. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 8 / Pre-Algebra

8.EE.B.6 – problems 3, 7, 8, 12

8.F.B – problems 1, 2, 4, 5, 6, 9, 10, 11

Difficulty 1–4 of 5 across 26 tagged checks.

1. Table $x = 0, 1, 2, 3$ against $y = 6, 10, 14, 18$. (a) y -intercept. (b) Slope.

Solution (a): The table already contains $x = 0$, so read its output directly.

$$b = 6$$

Solution (b): From the first column to the last, y rises $18 - 6 = 12$ while x rises $3 - 0 = 3$, so the slope is $12/3$. The equation is $y = 4x + 6$.

$$m = 4$$

2. Taxi: \$3 to start plus \$2 per mile. Cost of a 12-mile ride.

Solution: The \$3 is charged once, so it is the constant; the \$2 is charged per mile, so it multiplies the miles: $y = 2x + 3$. At $x = 12$: $2(12) + 3$.

$$27$$

3. Line P through $(0, 2)$ and $(4, 14)$. (a) Slope. (b) y at $x = 7$.

Solution (a): $\frac{14 - 2}{4 - 0} = \frac{12}{4}$.

$$m = 3$$

Solution (b): The marked point $(0, 2)$ is on the y -axis, so the equation is $y = 3x + 2$. At $x = 7$: $3(7) + 2$.

$$y = 23$$

4. Table $x = 0, 2, 4$ against $y = 4, 14, 24$, and the equation $y = 5x + 4$. (a) Table value at $x = 4$. (b) Equation value at $x = 4$. (c) Do they match?

Solution (a): Read the last column.

24

Solution (b): $5(4) + 4 = 20 + 4$.

24

Solution (c) — instructor-judged. They do match, but one agreeing value is not a proof. A full answer checks both numbers: the table gives 4 when x is 0, which is the constant in the equation, and it rises 10 over every 2 steps of x , which is a slope of 5 — the number multiplying x . Checking a second row is equally good evidence.

5. Battery: 100 percent, dropping 8 points per hour. Charge after 9 hours.

Solution: The starting 100 is the constant; the loss of 8 per hour multiplies the time and is subtracted: $y = 100 - 8t$. At $t = 9$: $100 - 72$.

28

6. Draining tank: minutes 0, 3, 6, 9 against litres 40, 31, 22, 13. (a) y -intercept. (b) Slope.

Solution (a): The column at 0 minutes.

$b = 40$

Solution (b): $\frac{13 - 40}{9 - 0} = \frac{-27}{9}$. The tank is losing water, so the slope is negative.

$m = -3$

7. Line Q through $(0, 10)$ and $(5, 0)$. (a) Slope. (b) y at $x = 3$.

Solution (a): $\frac{0 - 10}{5 - 0} = \frac{-10}{5}$.

$m = -2$

Solution (b): The equation is $y = 10 - 2x$. At $x = 3$: $10 - 6$.

$y = 4$

8. Line R through $(0, 1)$ and $(6, 13)$; a student offers $y = 3x + 1$. (a) Slope of R . (b) Equation value at $x = 6$. (c) Does it match?

Solution (a): $\frac{13 - 1}{6 - 0} = \frac{12}{6}$.

$m = 2$

Solution (b): $3(6) + 1 = 18 + 1$.

19

Solution (c) — instructor-judged. It does not match. The equation gives 19 at $x = 6$, but the marked point on line R is 13, and the slopes disagree: 3 in the equation against 2 on the graph. The constants happen to agree, which is exactly why one matching number proves nothing. A slope-only argument earns full credit provided both numbers are stated.

9. Gym: $C = 18m + 50$ dollars after m months. (a) Cost after 7 months. (b) Months to reach \$284.

Solution (a): $18(7) + 50 = 126 + 50$.

176

Solution (b): Solve $18m + 50 = 284$: subtract 50 to get $18m = 234$, then divide by 18.

$m = 13$

10. Table $x = 0, 4, 8, 12$ against $y = 9, 15, 21, 27$. (a) Change in y . (b) Slope.

Solution (a): $27 - 9$.

18

Solution (b): That rise happens over $12 - 0 = 12$ steps of x , so the slope is $\frac{18}{12}$, which reduces. Each single step of x raises y by one and a half.

$m = \frac{3}{2}$

11. Pool: 30 litres already in it, gaining 12 litres per minute; table 0, 2, 4 minutes against 30, 54, 78 litres; equation $L = 12t + 30$. (a) Table value at 4 minutes. (b) Equation value at 4 minutes. (c) Why all three agree.

Solution (a): Read the last column of the table.

78

Solution (b): $12(4) + 30 = 48 + 30$.

78

Solution (c) — instructor-judged. Both numbers appear in all three descriptions. The 30 litres already in the pool is the table's first entry and the constant in the equation; the 12 litres per minute is the 24-litre rise across every 2 minutes of the table and the number multiplying t . A correct answer names both correspondences, not just the one shared value.

12. Line P , the table $x = 0, 2, 4$ against $y = 2, 8, 14$, and the equation $y = 3x + 2$. (a) Table value at $x = 4$. (b) y at $x = 20$. (c) Why the equation can answer (b).

Solution (a): Read the last column.

14

Solution (b): $3(20) + 2 = 60 + 2$.

62

Solution (c) — instructor-judged. The table stops at $x = 4$ and the drawn line stops at the edge of the grid, so neither shows what happens at $x = 20$; the equation is a rule that accepts any input and returns the value directly. An answer that says the table could be continued or the line extended earns credit provided it also notes that the equation gives the value at once.